



Centre launches AI portal

The Indian Government has launched an AI portal to track AI related developments within the nation. <https://digit.in/jun20-96>



iPhone production increase?

A recent change in govt rules could mean a sharp increase in Apple production in India. <https://digit.in/jun20-97>

use in 50 days. Therefore, BERT became a rather obvious selection for Tiny AI researchers who wanted to shrink large AI models. Reportedly, Huazhong University of Science and Technology and the Huawei Noah's Ark Lab were successful at building TinyBERT. TinyBERT was 7.5 times smaller than the original BERT but was also 9.4 times faster than the original BERT too! According to the report by Huawei and Huazhong University, TinyBERT is almost just as accurate as BERT since it achieves 96 percent of the performance of the original BERT.

How are researchers going about achieving this, you ask? Tiny AI researchers develop 'knowledge distillation' methods that reduce the size of the AI model. With these distillation methods, an AI model can be scaled down to almost one tenth its size. These smaller AI models can be deployed on the 'edge' with inbuilt algorithms. This eliminates the need to send data over to the cloud to process and thereby reduces latency since the algorithms run on the device itself. Despite the reduced size, this method accelerates inference and maintains high levels of accuracy. This process involves training a smaller AI environment, let's call it the student,

from a larger, more sophisticated AI model, say the teacher. The training process involves running various iterations of data on both models in order to compare and tune the output of the student. Eventually, the student will be capable of producing almost the same/the same outcome as the teacher. This allows the development of a smaller AI engine that boasts the capabilities of a larger AI model.

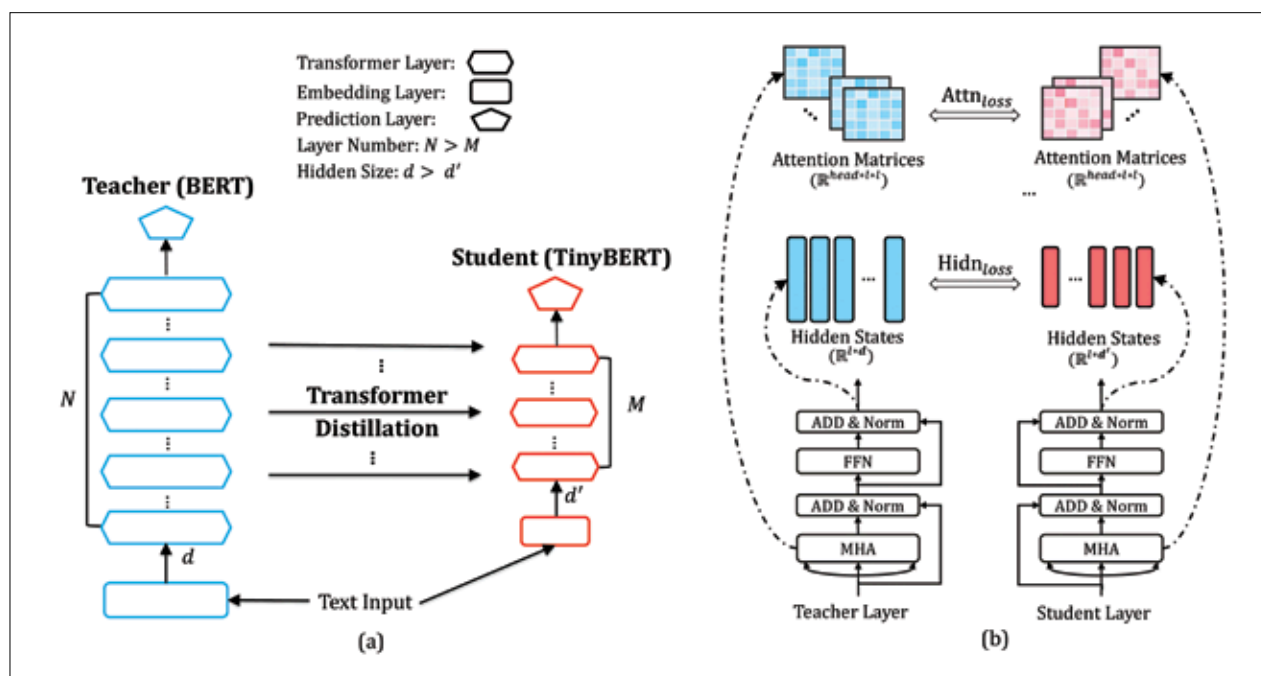
THE NEED FOR TINY AI

As we briefly discussed above, training sophisticated AI models consumes massive amounts of energy. AI adoption is increasing exponentially every passing day and the need to make the technology greener is apparent. Another factor that is facilitating the push for more power efficient AI is the need to run sophisticated AI models on smaller devices at the 'edge'. This will ultimately allow for advanced use cases in robotics, automated video security, voice assistants, autocorrection, image processing in cameras, autonomous driving, connected healthcare, Industry 4.0, precision farming, and many many more areas. Additionally, reducing the size of AI and machine learning models also reduces the need for massive amounts of computational

power. It lowers overall cloud and power costs. This could be a huge boon to the concerned industries since it is commonly seen that for every dollar spent on AI, there's an extra \$10-15 spent on cloud computing to support the infrastructure.

The AI of the future needs to be able to run on much smaller microprocessors, many of which are powered by batteries, such as smartphones, and a host of IoT devices. For example, mobile cameras can be used for medical image analysis or autonomous driving could be conducted without the cloud (which saves invaluable microseconds).

The ability to conduct sophisticated AI programs on small-form factor devices without the need to go back and forth with the cloud will allow these devices to be on the edge of the 'edge' - one step further than edge computing. Tiny AI involves building complex AI algorithms into the hardware at the very periphery of a network, in most cases, the devices or sensors themselves. By integrating these algorithms into the hardware, data analytics can be accurately performed at much lower power due to the absence of integration with the cloud. This also improves privacy manifold since the data never leaves the device and is therefore less



An overview of the distillation process used to create TinyBERT